

SAFDAR AYUB

Agentic AI Engineer | LLM Systems Developer | MCP & RAG Specialist

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PROFESSIONAL SUMMARY

Agentic AI Engineer with hands-on experience designing and deploying production-grade autonomous agent systems powered by **OpenAI Agents SDK**, **Claude Code (Opus)**, and **MCP Protocol**. Built a Hackathon Platinum-tier AI agent with **4 custom MCP servers**, **event-driven triggers**, **circuit breaker patterns**, and **human-in-the-loop approval gates** — monitoring Gmail, filesystem, WhatsApp, and Odoo ERP. Implemented **RAG pipelines** using Qdrant Cloud vector database and OpenAI embeddings. Proficient in **Python**, **FastAPI**, **prompt engineering**, **LLM orchestration**, and **REST API** integration. Brings military-grade discipline — 23+ years ensuring zero-defect, mission-critical systems — to AI reliability and production deployment.

TECHNICAL SKILLS

Agentic AI Frameworks	OpenAI Agents SDK, MCP SDK, FastMCP (4 custom servers), Claude Code (Opus) — preferred over LangChain for production reliability and type safety
LLM Integration	OpenAI APIs, Claude API (Anthropic), Groq API, Gemini TTS — prompt engineering, chaining strategies, tool use, structured outputs
RAG & Vector DBs	Qdrant Cloud, RAG pipeline design, semantic search, document chunking, embedding strategies — analogous to Pinecone/Weaviate
AI Workflow Design	Priority-queue orchestration, human-in-the-loop gates, event-driven triggers, circuit breaker patterns, risk-based routing
Backend & APIs	Python, FastAPI, Node.js, REST APIs, Webhooks, APScheduler, PM2 process management, Playwright automation
Databases	PostgreSQL (Neon), SQLite, Turso/LibSQL, Prisma, SQLAlchemy, Qdrant Cloud, Upstash Redis, Supabase
Frontend & Full Stack	Next.js 16, React, TypeScript, Tailwind CSS, ShadCN UI, Recharts, Zustand
Cloud & DevOps	Docker, Kubernetes, Helm Charts, OCI/OKE (Oracle Cloud), NGINX Ingress, Dapr, Apache Kafka, GitHub Actions, Vercel
Auth & Integrations	Gmail OAuth2, Google OAuth, NextAuth.js v5, Better Auth, JWT, Odoo RPC, Stripe, Resend, Cloudinary

KEY PROJECTS

Personal AI Employee — Autonomous Multi-Agent System

HACKATHON PLATINUM

GitHub

MOST RELEVANT

Production-grade autonomous agent system — event-driven, multi-tool, human-in-the-loop, 4 custom MCP servers

- Designed and built local-first AI agent monitoring **Gmail**, **filesystem**, and **WhatsApp** with **priority-queue orchestration** and human-in-the-loop approval gates for sensitive/critical actions
- Implemented **4 custom FastMCP servers** (email, social media, Odoo ERP, documents) with **circuit breaker patterns** and correlation ID tracking — production-reliability patterns for autonomous systems
- Designed **risk-based action routing** (routine / sensitive / critical) with dry-run defaults, ensuring safe autonomous operation without human oversight failures
- Integrated **Obsidian vault pipeline** with 12 Claude Code skills for automated task processing and weekly CEO briefing generation — end-to-end AI workflow automation
- Managed multi-process orchestration with **PM2** and **APScheduler** for production-grade reliability and scheduling

Stack: Python, FastMCP, Claude Code (Opus), OpenAI Agents SDK, Gmail OAuth2, Playwright, Odoo RPC, APScheduler, PM2

Physical AI & Humanoid Robotics Interactive Textbook — RAG Chatbot

GitHub | Live

Production RAG system — OpenAI Agents SDK + Qdrant vector database + FastAPI backend

- Built **RAG chatbot** using **OpenAI Agents SDK**, FastAPI, Neon Postgres, and **Qdrant Cloud** vector database — semantic search over 6-chapter curriculum content
- Implemented document chunking, embedding pipeline, and semantic retrieval — full RAG architecture from ingestion to response
- Integrated **MCP SDK** for tool-use extensions and user authentication (Better Auth) with role-based access

Stack: Docusaurus, React, FastAPI, OpenAI Agents SDK, MCP SDK, Neon DB, Qdrant Cloud, Better Auth, Vercel

AI Video Generation Agent — End-to-End LLM Orchestration Pipeline

[GitHub](#) | [YouTube Demo](#)

Multi-step LLM pipeline — prompt chaining, tool orchestration, TTS integration, video rendering

- Built AI pipeline that generates complete motion graphics videos from text prompts: **script** → **storyboard** → **scenes** → **narration** → **rendered video** — demonstrating advanced LLM chaining strategies
- Used **Claude Code (Opus)** as **LLM orchestrator** with Remotion/React rendering engine and **Gemini TTS** for synchronized voiceover — multi-model AI workflow
- Produced 4-minute 1080p explainer video with spring physics animations — validates AI system output quality end-to-end

Stack: TypeScript, React, Remotion, Claude Code (Opus), Gemini TTS, Spring Physics

AI-Powered Todo Application — 8-Phase Cloud-Native Evolution

[GitHub](#) | [Live](#)

- Integrated **Groq API** LLM for natural language task management with conversation history — production AI chatbot within a full-stack app
- Evolved across 8 phases to **Kubernetes on OCI OKE** with Helm charts, Dapr sidecar injection, and Apache Kafka events — demonstrates AI system scalability and cloud deployment

Stack: Next.js, FastAPI, SQLAlchemy, Groq API, Docker, Kubernetes, Helm 3, Dapr, Apache Kafka, OCI OKE

PROFESSIONAL EXPERIENCE

Quality Control Inspector & Electronics Technician (Retired)

Feb 2000 – Mar 2023 (23 Years)

Pakistan Air Force

- Maintained mission-critical communication and avionics systems under strict operational SLAs — discipline directly applied to AI system reliability, monitoring, and zero-downtime standards
- Led quality control inspection enforcing zero-defect standards — directly maps to AI output quality assurance and response monitoring
- Managed cross-functional technical teams — mirrors collaboration with product, engineering, and data teams in AI delivery

EDUCATION & CERTIFICATIONS

Certified Agentic and Robotic AI Engineer — PIAIC, Air University Islamabad

Expected 2026

Master of Computer Science (CGPA: 3.11) — Virtual University of Pakistan

2012

DAE Radio Technology (71%) — School of Electronics, PAF Korangi Creek

2003

Artificial Intelligence & Computing Programme — PIAIC Presidential Initiative (2023) | PCAP: Programming Essentials in Python — Cisco / OpenEDG (2022)

Portfolio & Demos: [safdarayub-dev.vercel.app](#) | GitHub: [github.com/safdarayubpk](#) | Personal AI Employee demonstrates production-grade agentic AI with MCP servers, circuit breakers, and multi-tool orchestration.